AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII EXAMINATION

APRIL/ MAY 2017

Business Statistics Paper II

Time: 1 hour 30 minutes Marks: 30

AKU May Examination 2017

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are SEVEN questions. Answer ALL questions. Choices are specified inside the paper.

3. When answering the questions:

   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a scientific calculator if you wish.

Q.1. (Total 4 Marks)

i. Define continuous variable and give ONE example of it. (2 Marks)

ii. Write any TWO characteristics of a good questionnaire. (2 Marks)

Q.2. (Total 4 Marks)

Mr Amjad asked the students of his class about their favourite ice cream flavour. The students' responses are given in the following grid.

Mango	Vanilla	Mango	Chocolate	Chocolate
Vanilla	Mango	Chocolate	Chocolate	Chocolate
Tutti-frutti	Vanilla	Chocolate	Mango	Tutti-frutti
Chocolate	Tutti-frutti	Chocolate	Tutti-frutti	Tutti-frutti
Tutti-frutti	Vanilla	Mango	Chocolate	Mango

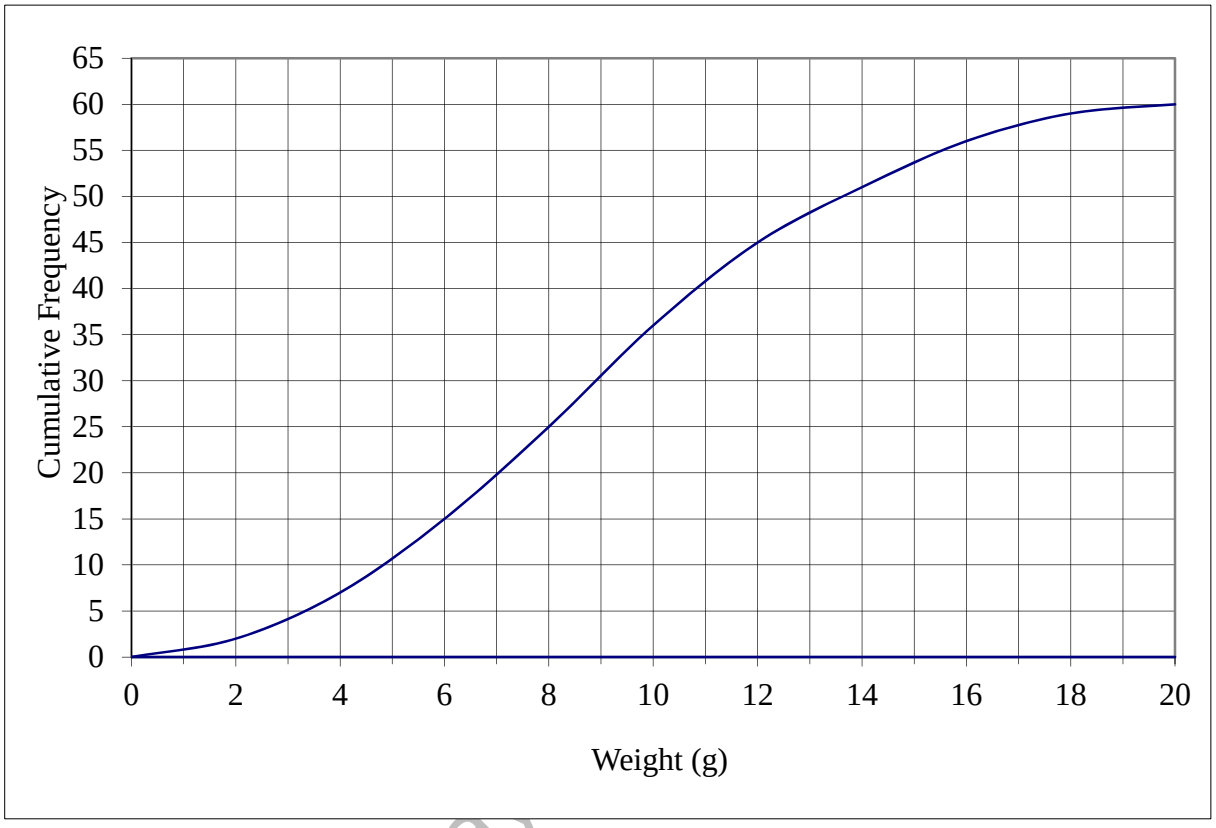
Fill the given table to construct the frequency distribution of the data given in the grid. Also find the relative frequency for each flavour.

Flavours	Tally Marks	Frequency	Relative Frequency

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Q.3. (Total 4 Marks)

The given cumulative frequency curve shows the weight (in grams) of 60 packages to be delivered by a courier company.



Using the given cumulative frequency curve, estimate the

i. number of packages that weigh between 6 grams to 12 grams. (2 Marks)

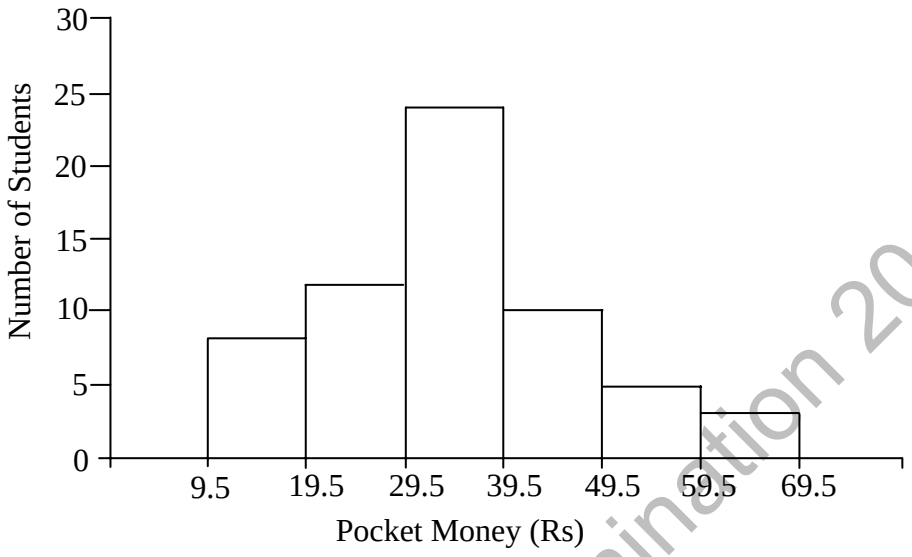
ii. median weight (in grams) of the packages. (2 Marks)

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(**ATTEMPT EITHER PART a OR PART b OF Q.4.)**

Q.4. (Total 5 Marks)

a. The diagram shows a histogram which represents the daily pocket money (in Rupees) of students in a class.



i. Estimate the mode of this data graphically. (1 Mark)

ii. The frequency distribution of this histogram is shown in the given table. Fill in the table appropriately to find the mean of the data. (2 Marks)

Class Boundary	**Number of Students**	**Class Mark**	
9.5 – 19.5	9	14.5	
19.5 – 29.5	12	24.5	
29.5 – 39.5	24	34.5	
39.5 – 49.5	10	44.5	
49.5 – 59.5	5	54.5	
59.5 – 69.5	3	64.5	
Total	63	—	

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iii. Using the mode and mean of the data found in part (i) and (ii), estimate the median of the data. (2 Marks)

(**Hint:** Use empirical relationship)

b. The ages (in years) of a sample of 31 students were recorded. The results are shown in the following table. (5 Marks)

Age (years)	Number of Students	Class Boundary
6 – 9	7	
10 – 13	8	
14 – 17	6	
18 – 21	10	
Total	31	—

Find the mode of this data.

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Q.5. (Total 4 Marks)

A teacher designed a class project which was to be done by the entire class in groups of 5. Students of Group *A* managed to obtain full marks in the class project. After a certain time, the teacher sets up a 3-hour class test for the entire class and announced that the marks they obtain in this test will be added to the marks they had previously obtained in their class project.

The test marks obtained by the students of group *A* are 75, 88, 96, 52, 61.

i. Find the standard deviation of the test marks of group *A*. (2 Marks)

ii. Each student of group *A* obtained 10 marks for the class project which will be added to their test marks. Would the standard deviation of the total marks after adding class project marks of group *A* be different from the standard deviation of the original test marks of this group? Support your answer with suitable property of measures of dispersion. (2 Marks)

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(ATTEMPT EITHER PART a OR PART b OF Q.6.)

Q.6. (Total 5 Marks)

a. In the given table, the prices (in Rupees per kilograms) and quantities (in kilograms) of some commodities in the years 2006 and 2007 are given. Find Laspeyer's index number, taking year 2006 as the base period.

Commodities	2006		2007			
	Price (Rs)	Quantity (kg)	Price (Rs)	Quantity (kg)		
Rice	75	15	80	12		
Chick Pea	55	6	65	5		
Flour	28	20	30	20		
Sugar	50	8	60	7		
Total						

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(ATTEMPT EITHER PART a OR PART b OF Q.6.)

b. In the given table, the prices (in Rupees per kilograms) and quantities (in kilograms) of three commodities *X*, *Y* and *Z* in the years 2008 and 2009 are given. Find Fisher's index number for the following data, taking year 2008 as the base period.

Commodities	**Prices (Rs/ kg)**		**Quantities (kg)**			
	2008	**2009**	**2008**	**2009**		
X	16	20	40	50		
Y	12	15	90	100		
Z	14	16	50	60		
Total						

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(ATTEMPT EITHER PART a OR PART b OF Q.7.)

Q.7. (Total 4 Marks)

a. Basit and Adnan both have a fair die with faces numbered 1, 2, 3, 4, 5 and 6. They throw their dice simultaneously.

(**Note:** Singular: die, Plural: dice)

i. Write down the sample space. (2 Marks)

Expressing each answer as a fraction in its lowest term, find the probability that they both get

ii. an odd number. (1 Mark)

iii. number 9. (1 Mark)

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(**ATTEMPT EITHER PART a OR PART b OF Q.7.)**

b. A box contains 15 blue balls, 12 red balls and 23 black balls. If Sara draws a ball at random from this box, find the probability it is a/ an

i. black ball. (1 Mark)

ii. blue or red ball. (2 Marks)

iii. orange ball. (1 Mark)

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